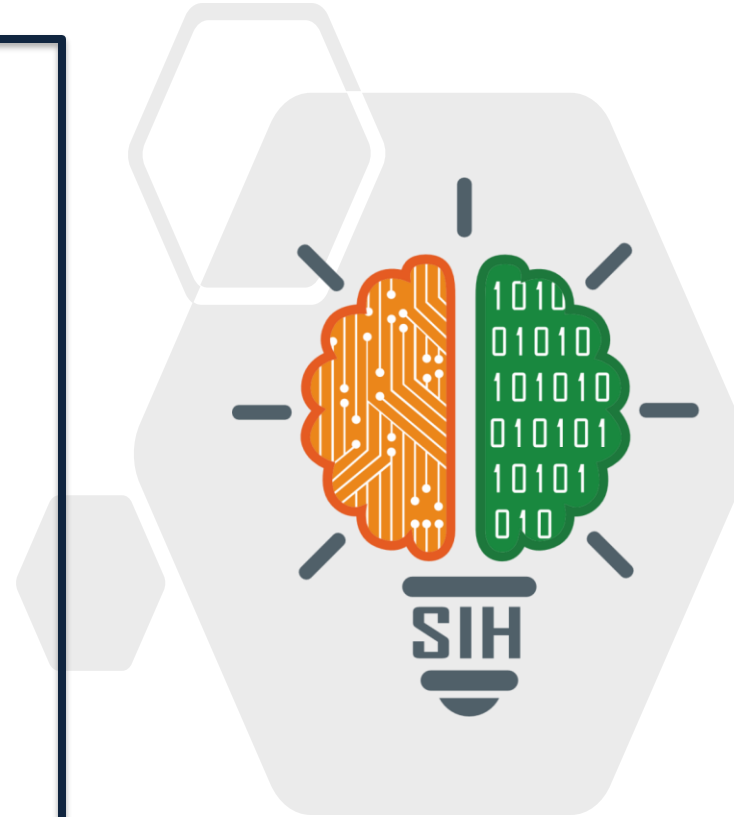




- Problem Statement ID : SIH25067
- Problem Statement Title : Proposal for Design and Development of Application for Heavy Metal Pollution Indices
- Theme : Miscellaneous
- PS Category : Software





CAUSE : Historical chromium-bearing industrial waste associated with former TCCL site.



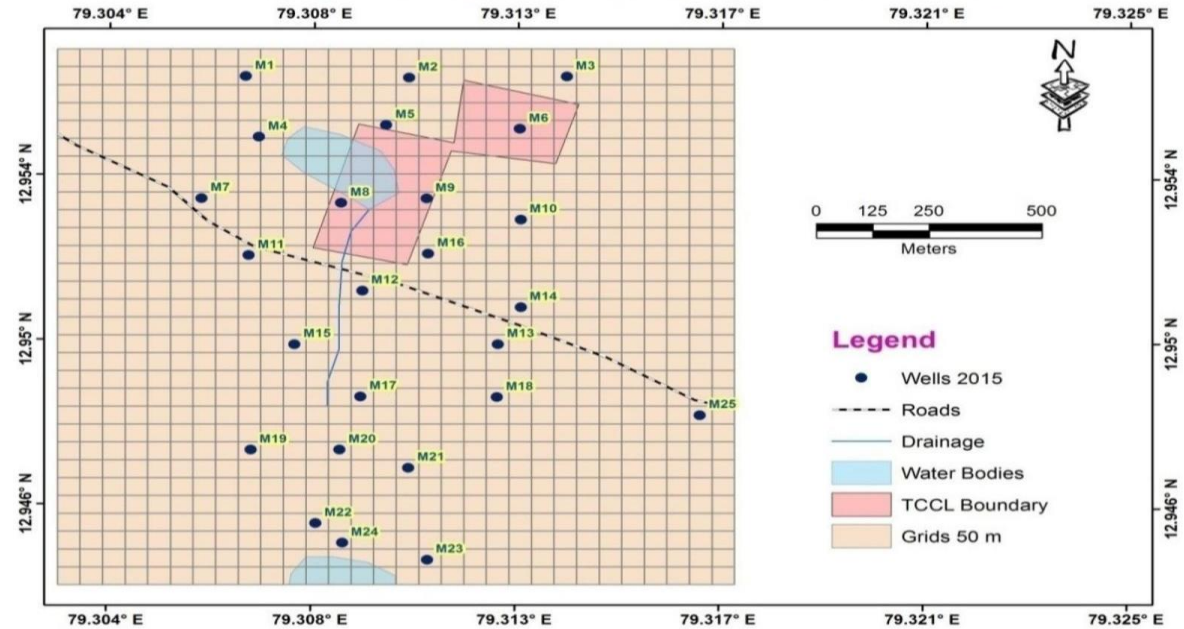
IMPACT : Chromium contamination affected surrounding soil and groundwater.



EXTENT : Contamination was reported to extend approximately 2 km south of the site.



AFFECTED : Nearby communities, residential areas and agricultural water resources.



WHAT THE MAP SHOWS:

- **TCCL Boundary** – Historical waste site
- **Blue Points** – Groundwater monitoring wells
- **Water Bodies & Drainage** – Local water pathways
- **Study Area** – Groundwater monitoring region



Intelligent Groundwater Heavy-Metal Assessment & Decision-Support Platform

OUR SOLUTION-END TO END WORK FLOW

01



DATA INGESTION :

- Manual Entry / CSV Upload
- Metal concentrations + coordinates + date

02



VALIDATE & STANDARDIZE :

- Detect missing values & unit issues
- Anomaly detection & data quality check

03



POLLUTION INDEX ENGINE :

- Compute HPI, HEI, Cd, MI, Igeo, PLI
- BIS / WHO / Custom standards

04



HEALTH RISK ASSESSMENT :

- HQ, HI & Carcinogenic Risk
- Separate for Adults & Children

05



LOCATION & RADIUS ANALYSIS :

- Select any location & radius
- Nearby samples, hotspots & sampling gaps

06



SOURCE ATTRIBUTION :

- Evaluate probable source of contamination
- Geogenic / Anthropogenic / Mixed

07



INTERVENTION PRIORITIZATION :

- Rank locations using pollution severity, health risk & exposure factors

LOCATION-BASED INTELLIGENCE MOCKUP



TECH STACK TO BE USED



Frontend

React TypeScript Vite Tailwind CSS Shadcn/ui



Backend

Python FastAPI Pydantic Uvicorn



Data science

Pandas Numpy Scipy Scikit-learn



GIS & MAPPING

GeoPandas Shapely PostGIS React Leaflet
IDW OSM GeoJSON

PostgreSQL



Database

PostgreSQL PostGIS



Reports & visualization

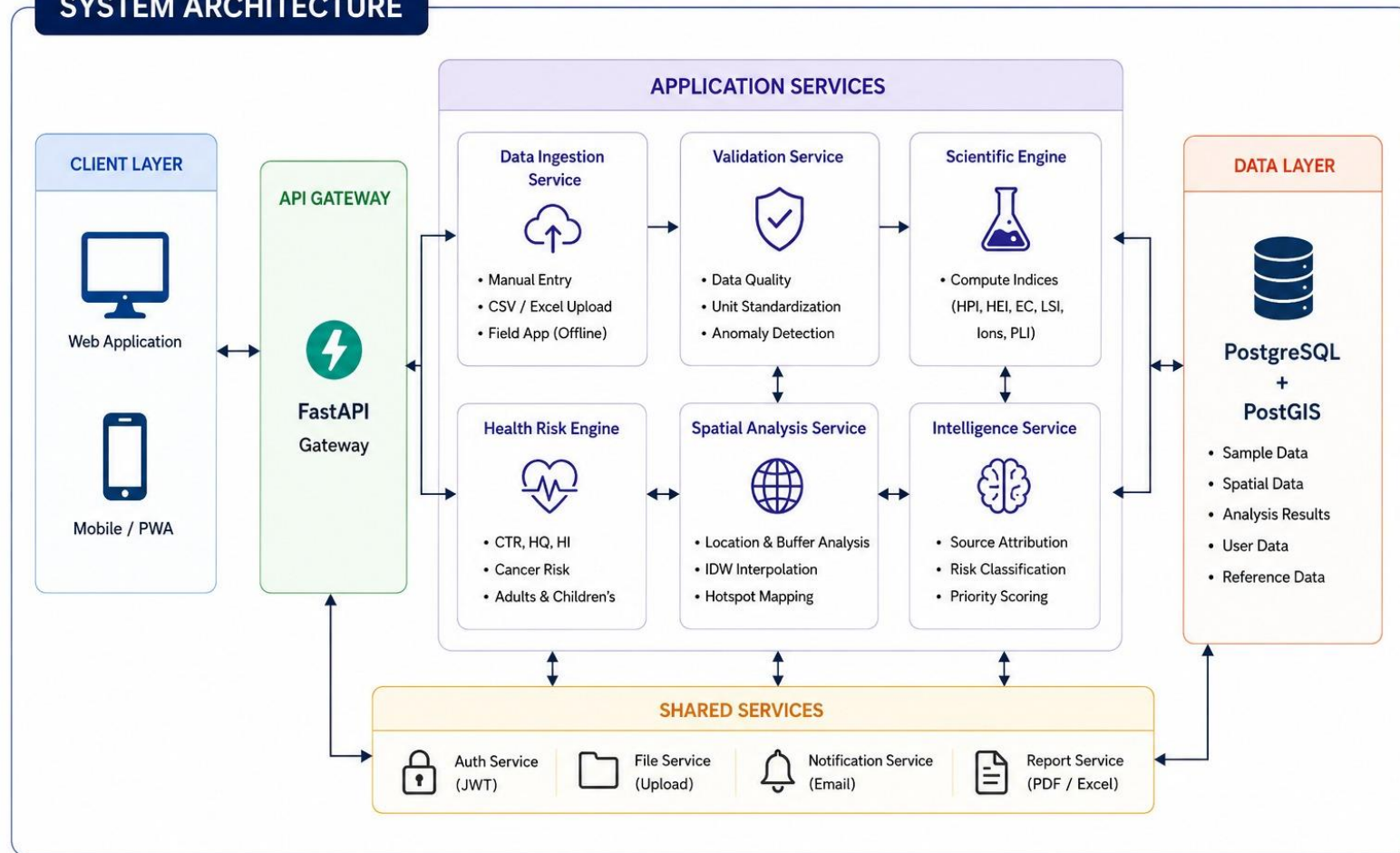
Recharts ReportLab Interactive Dashboard



Deployment

Docker AWS

SYSTEM ARCHITECTURE



SECURITY & ACCESS

- JWT Authentication
- Role-based Access Control
- Secure REST APIs
- Data Integrity & Privacy

← Assessing Practicality. Ensuring Impact. →

FEASIBILITY AT A GLANCE

TECHNICAL FEASIBILITY

- Built on open-source stack
- AI/ML models run efficiently on low-end devices & cloud

FINANCIAL FEASIBILITY

- Low-cost development using open-sourcetools
- Scalable via cloud & subscription model

OPERATIONAL FEASIBILITY

- PWA enables offline data collection & sync
- Simple UI for field teams, automated alerts& reports

SCALABILITY & SUSTAINABILITY

- Scales from sample points to state-widecoverage
- Modular design for easy integration &expansion

PILOT TO STATE-WIDE IMPACT



SAMPLE DATA

Collect groundwater samples & parameters



PILOT AREA

Validate models & workflows in real field



BLOCK LEVEL

Expand to multiple locations & analyze trends



DISTRICT LEVEL

Generate actionable insights for decision makers



STATE-WIDE

Unified dashboard for monitoring & policy action

CHALLENGES

OUR MITIGATION



MESSY & INCOMPLETE DATA



Smart validation + anomaly Detection



LOW CONNECTIVITY IN FIELD AREAS



online PWA + auto-sync



VARIATION IN STANDARDS & REGULATIONS



BIS / WHO / Custom thresholds



UNCERTAINTY IN SOURCE & RISK ATTRIBUTION



Source attribution + priority scoring



GROUND WATER DECISION-SUPPORT PLATFORM

1.SCIENTIFIC ASSESSMENT

Multi-index pollution and health-risk analysis for reliable assessment

2.SPATIAL INTELLIGENCE

Converts sample data into contamination hotspots and district level insights

3.ACTIONABLE MONITORING

Identifies probable sources and prioritizes area for investigation and remediation

TARGETED BENEFICIARIES

- 1.Pollution Control Boards.
- 2.PHED / Jal Jeevan Mission authorities
- 3.Scientists & researchers
- 4.Groundwater-dependent communities



IMPACT AND BENEFITS

SOCIAL BENEFITS

PUBLIC HEALTH PROTECTION

Adult and child health-risk assessment highlights potential high-risk exposure zones



COMMUNITY AWARENESS

Clear village-level water quality Information improves community awareness and safety



GOVERNANCE BENEFITS

EFFICIENT MONITORING

Reduces manual analysis, processing and report time.



TARGETED RESOURCE ALLOCATION

Directs testing, investigation and remediation resources to priority locations



IMPACT OF OUR PLATFORM



Stronger Scientific Decisions

Multiple validated assessments give authorities greater confidence before taking action.



Faster Response to Critical Areas

Authorities can focus limited testing and remediation efforts where they are most needed.



Reduced Exposure to Unsafe Water

Earlier identification of concerning locations enables timely investigation and protective measures.



Long-Term Groundwater Protection

Continuous, location-based insights support better planning and sustainable groundwater management.

1



01 | BIS — IS 10500:2012 Drinking Water — Specification

https://www.bis.gov.in/other/DrinWat/S10500.pdf?utm_source=chatgpt.com

2



02 | WHO — Guidelines for Drinking-water Quality

https://www.who.int/publications/i/item/9789240121225?utm_source=chatgpt.com

3



03 | CGWB — Central Ground Water Board

https://cgwb.gov.in/ground-water-quality?utm_source=chatgpt.com

4



04 | India-WRIS

<https://indiawris.gov.in/>

KEY RESEARCH REFERENCES

- **Sharma et al. (2022)** — HPI, HEI, Cd & health-risk assessment
[Research Paper](#)
- **Chorol & Gupta (2023)** — HPI & Monte Carlo health-risk assessment
[Research Paper](#)
- **Kadaladi, Tamil Nadu (2025)** — HPI, MI, HQ & GIS hotspot analysis
[Research Paper](#)
- **Srivaikundam, Southern India (2025)** — HPI, HEI, HQ, HI, CR & GIS
[Research Paper](#)



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